

Supplemental Material: Yukawa screening derivation of the bond-valence rule

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I. DATA PROVENANCE

The figures and tables in this Supplemental Material cover the same 103 cation–oxygen species analyzed in the main text, spanning *s*-, *p*-, *d*-, and *f*-block cations.

Source structures are DFT-relaxed entries from the Materials Project, retrieved with an energy-above-hull window of 0 to 50 meV/atom, so every structure lies within 50 meV/atom of the convex hull. For each structure, bond strengths were obtained by solving the valence-sum and loop constraints on the cation–oxygen bond graph, and (R_0, B) were determined by linear least squares. For rank-deficient bond graphs (fewer independent bonds than parameters), an anion-centered reparameterization was used. Any remaining ill-conditioned cases were resolved with the constrained nonlinear procedure of Ref. 1, with R_0 bounds $(-10, 10)$ Å. Coordination numbers were determined from the bond graph.

The DFT-centroid comparison in Fig. 2 of the main text uses ten *s*-block oxides in panel a (Li_2O , Na_2O , K_2O , Rb_2O , Cs_2O , BeO , MgO , CaO , SrO , and BaO) and 25 non-*s*-block oxides in panel b: Al_2O_3 , SiO_2 , GeO_2 , SnO_2 , PbO , TiO_2 , V_2O_5 , CoO , NiO , CuO , ZnO , ZrO_2 , MoO_3 , PdO , Ag_2O , HfO_2 , Ta_2O_5 , Au_2O_3 , La_2O_3 , Gd_2O_3 , Yb_2O_3 , B_2O_3 , P_2O_5 , As_2O_3 , and Sb_2O_3 . The panel-*b* regression is defined a priori as a test of the Yukawa screening picture only for ground-state binary oxides whose cation environments approximate an isotropic first-shell ionic screening cloud. It therefore uses the 21 filled-symbol oxides and excludes B_2O_3 , P_2O_5 , As_2O_3 , and Au_2O_3 , whose ground-state structures are dominated by directional, non-isotropic bonding motifs. Those four compounds are shown as open symbols. The 21 oxides shown as filled symbols in main-text Fig. 2(b) are the 25 listed above minus the four directional cases just named. No further data-driven exclusion was applied beyond this fixed list. The excluded oxides are the cases in this benchmark whose ground-state cation environments are not compact first-shell ionic polyhedra but directional or terminal-oxyanion motifs, so a single shell centroid m_i is not an adequate descriptor of the screening cloud.

II. COORDINATION-NUMBER-RESOLVED B – R_0 FIT LINES

Figures S2–S16 show the n -resolved B vs. R_0 fit lines for every species in the 103-cation set, grouped by block: Group 1 (Fig. S2), Group 2 (Fig. S3), $3d/4d/5d$ transition metals (Figs. S4–S11), *p*-block (Figs. S12–S14), and *f*-block lanthanides plus actinides (Figs. S15–S16). Each caption quotes the species panels of its page in reading order, either in full or as a first–last range. Within each block, panels are ordered alphabetically by element symbol and oxidation state. Each panel shows the per-structure (R_0, B) pairs beneath the fitted lines, as filled circles for the RANSAC inliers of each line (in the line’s color) and as open circles for the RANSAC outliers. Axes are fixed to the window $0 < R_0 < 4$ Å and $-1 < B < 2$ Å, and the annotation above each panel gives the number of inlier structures that fall outside this window together with the number of RANSAC outliers. Inliers outside the window are nearly degenerate shells whose joint fit is weakly constrained along the line, so their R_0 and B can reach 10^2 – 10^3 Å while still lying on it. The fitted lines are drawn over the full R_0 range of their inliers and are clipped by the window. Within each coordination shell, the fitted parameters lie on a linear family $B = \beta^{(n)}R_0 + \beta_0^{(n)}$, and the lines converge toward cation-specific characteristic pairs (R_0^*, B^*) as described in the main text. Together, these pages display every coordination-resolved fit that contributes to the pole form $\beta = 1/\ln(z/n)$ shown in Fig. 1 of the main text.

Line-fitting procedure. Each DFT-relaxed structure contributes one independently fitted (R_0, B) pair. With the solved S_{ij} as data, the model $R_{ij} = R_0 - B \ln S_{ij}$ is linear in both parameters, which are determined jointly by ordinary least squares (OLS) over the structure’s cation–oxygen bonds. R_0 is fixed and B re-optimized only in the rare bound-projection fallback, when the unconstrained R_0 falls outside the pipeline’s R_0 bounds. A structure enters coordination class n only when its cation sites share an unambiguous oxygen coordination number in the bond graph. Mixed-coordination structures are left unbinned. For each species and coordination class with at least five structures, the line $B = \beta^{(n)}R_0 + \beta_0^{(n)}$ is fitted both by OLS and by random sample consensus (RANSAC), with residual threshold equal to twice the median absolute OLS residual and floored at 0.01 Å, and the RANSAC line

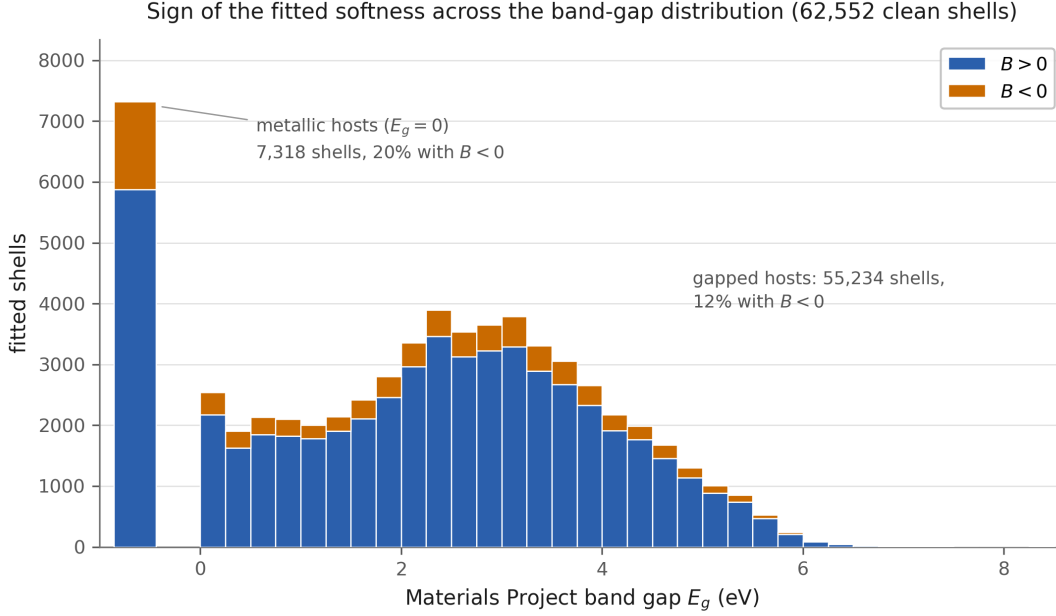


FIG. S1. Sign of the fitted softness B across the band-gap distribution of the corpus. Stacked counts of clean fitted shells (single-valence records with unregularized linear fits) versus the Materials Project band gap of the host structure, with metallic hosts ($E_g = 0$) drawn as the separate bar at left. The $B < 0$ fraction is 20% in metallic hosts and 12% across gapped hosts, consistent with over-screening by free carriers. Generated by `analysis/scripts/make_gap_sign_census_figure.py` from the band-gap map in `data/processed/theory/mp_band_gaps.json`.

is used whenever available. The reported slope and intercept come from an OLS refit on the RANSAC inlier set, while their standard errors and confidence intervals come from a 400-resample bootstrap over that inlier set. The population N_n of a line is the number of inlier structures in this final refit. It supplies the weights $w = \max(N_n, 1)$ used in the goodness-of-fit metrics of main-text Fig. 1 and in the characteristic-pair intersections below. Lines enter Fig. 1 of the main text only if they retain at least four inliers, have slope standard error at most 5, and have $R^2 \geq 0.3$.

Sign census of the per-shell fits. Of the 79,009 fitted shells in the corpus, 63,783 carry a single-valence oxidation-state assignment together with defined B , z , and n , and 16.4% of these return $B < 0$ (the main text's count of 7,565 shells at $z = n$ additionally includes mixed-valence records). The $B < 0$ fraction is 20.4% for $z < n$, 5.8% for $z = n$, and 10.2% for $z > n$. The negative sign is dominated by near-degenerate shells whose fitted B is barely constrained. The median bond-length range of the $B < 0$ shells is 0.059 Å, roughly half the 0.111 Å of the $B > 0$ shells, and 47% of $B < 0$ shells have ranges within 0.05 Å compared with 29% of $B > 0$ shells. The chemically systematic $B < 0$ fractions concentrate in square-planar and linear late transition metals (47% for Pd^{2+} , 43% for Au^{3+} , 37% for Cu^{1+}) and in large, soft cations in high, irregular coordination (51% for Eu^{2+} , 40% for Rb^+ , 31% for Sr^{2+}), consistent with directional bonding or ambiguous coordination producing anisotropic flux capture. Resolved against the band gap of the host structure, the $B < 0$ fraction is 20% among clean fits in metallic hosts versus 12% across gapped hosts (Fig. S1), consistent with over-screening where free carriers are present. The census is reproduced by `analysis/scripts/bv_sign_census.py` in the project repository [2].

Weighting scheme for the goodness-of-fit metrics. The weighted R^2 and mean absolute error reported with Fig. 1 of the main text use weights $w_i = \max(N_i, 1)$, where N_i is the number of DFT-relaxed structures retained as inliers in the final linear B - R_0 fit at species i and coordination n (see the line-fitting procedure above). This gives more populous (species, n) cells more leverage, in proportion to the number of independent structures backing each fitted slope β_i . Letting $\hat{\beta}_i = 1/\ln(z_i/n)$ be the parameter-free closed-form prediction and $\bar{\beta}_w = \sum_i w_i \beta_i / \sum_i w_i$, the reported metrics are

$$R_w^2 = 1 - \frac{\sum_i w_i (\beta_i - \hat{\beta}_i)^2}{\sum_i w_i (\beta_i - \bar{\beta}_w)^2}, \quad \text{MAE}_w = \frac{\sum_i w_i |\beta_i - \hat{\beta}_i|}{\sum_i w_i}.$$

Pooling 68 species at $n = 4$ ($\sum w = 13,370$) with 82 species at $n = 6$ ($\sum w = 17,162$) – the $z \neq n$ subset shown in Fig. 1 – gives $R_w^2 = 0.986$ and $\text{MAE}_w = 0.15$ for the parameter-free closed form $\beta = 1/\ln(z/n)$. The $n = 4$ panel alone gives $R_w^2 = 0.973$ with $\text{MAE}_w = 0.23$, and the $n = 6$ panel gives $R_w^2 = 0.979$ with $\text{MAE}_w = 0.089$.

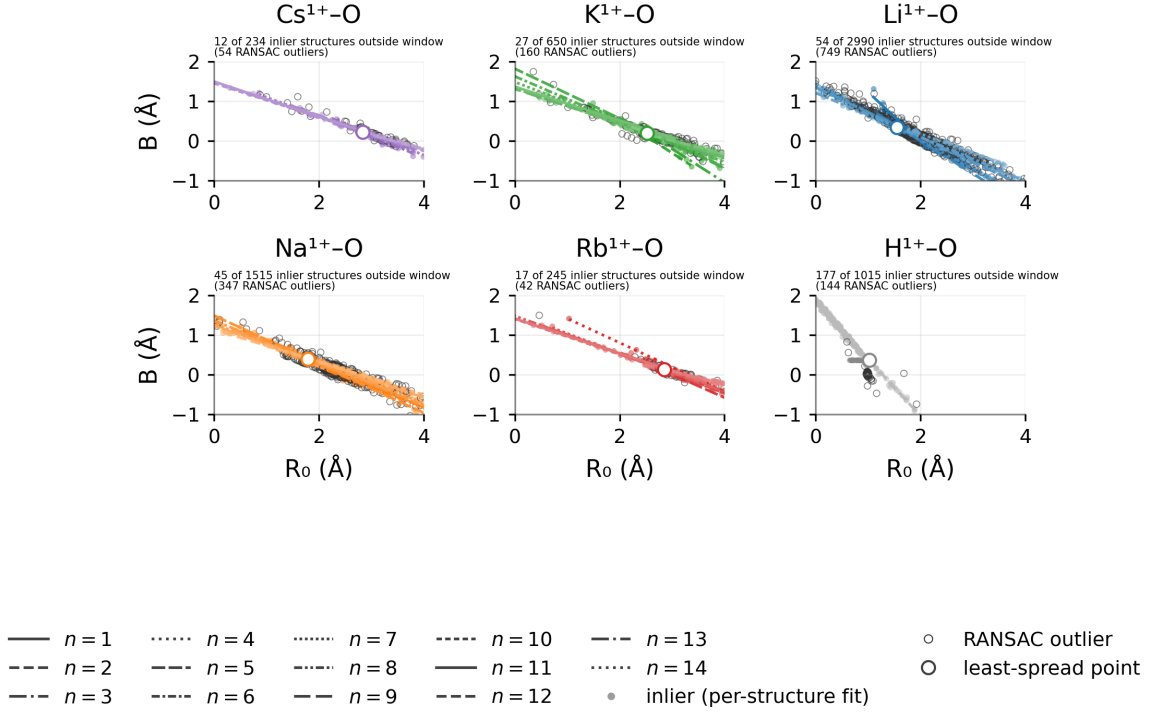


FIG. S2. n -resolved B vs. R_0 fit lines for Group 1 X–O pairs (Cs, K, Li, Na, Rb, H). Filled circles are the per-structure (R_0, B) inliers of each line, open circles the RANSAC outliers, line styles distinguish coordination numbers n (legend), and the large open circle marks the least-spread intersection point. Axes are fixed to $0 < R_0 < 4$ Å and $-1 < B < 2$ Å, with the number of inlier structures outside the window given above each panel. The convergence of the lines defines the characteristic pair (R_0^*, B^*) for each cation.

Slope standard errors. Each fitted slope carries a standard error σ_β from the 400-resample bootstrap described in the line-fitting procedure above. Across the 150 slopes the median relative standard error is 0.19% and the 90th percentile is 3.9%, roughly two orders of magnitude below the across-species spread at the formal charges flanking the pole. The per-species slopes are drawn with these error bars in Fig. S17.

Exclusion of $z = n$ species from main-text Fig. 1. Species at $z = n$ (U^{4+} , Mn^{4+} , Ti^{4+} , V^{4+} , Zr^{4+} , Ir^{4+} , P^{4+} , Pb^{4+} , and Te^{4+} for $n = 4$, and Mo^{6+} and W^{6+} for $n = 6$) are omitted from Fig. 1 of the main text. At $z = n$ the valence-sum rule fixes the mean bond valence at unity but not each S_{ij} individually. For a degenerate shell with equal bond lengths the rule is satisfied for any B with $R_0 = \bar{R}$, leaving the B – R_0 slope β formally unconstrained. For nearly degenerate shells the S_{ij} differ and B is finite, but the fit is ill-conditioned because the intercept ceases to constrain the softness at $z = n$ (see the screening-collapse discussion in the main text). Accordingly, the fits exhibit very large variance (see the $z = n$ panels in Figs. S4–S11 and S12–S16, where the coordination-resolved lines have widely scattered slopes and intercepts). The closed-form expression $\beta = 1/\ln(z/n)$ correctly diverges at $z = n$, and the divergence itself is a quantitative prediction of the Yukawa screening picture. We plot only $z \neq n$ species so that finite, well-conditioned fitted slopes are not visually conflated with the formal pole. Main-text Fig. 1 therefore shows the 94 species (150 fitted slopes) for which $z \neq n$ at $n = 4$ or $n = 6$. Six species (Cl^{1-} , C^{4+} , Hg^{1+} , H^{1+} , N^{3+} , S^{4+}) lack a fittable line at either $n = 4$ or $n = 6$, P^{4+} lies exactly at the $n = 4$ pole, and Te^{6+} and U^{6+} fall below the $R^2 \geq 0.3$ acceptance threshold at $n = 6$.

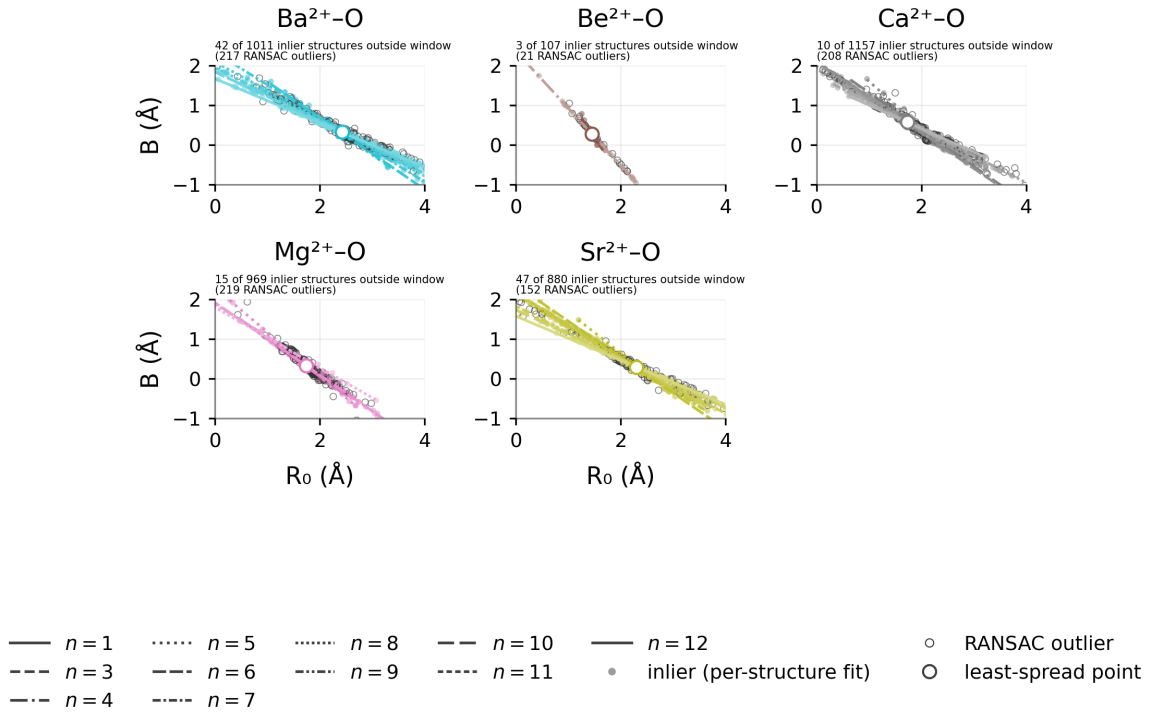


FIG. S3. n -resolved B vs. R_0 fit lines for Group 2 X-O pairs (Ba, Be, Ca, Mg, Sr).

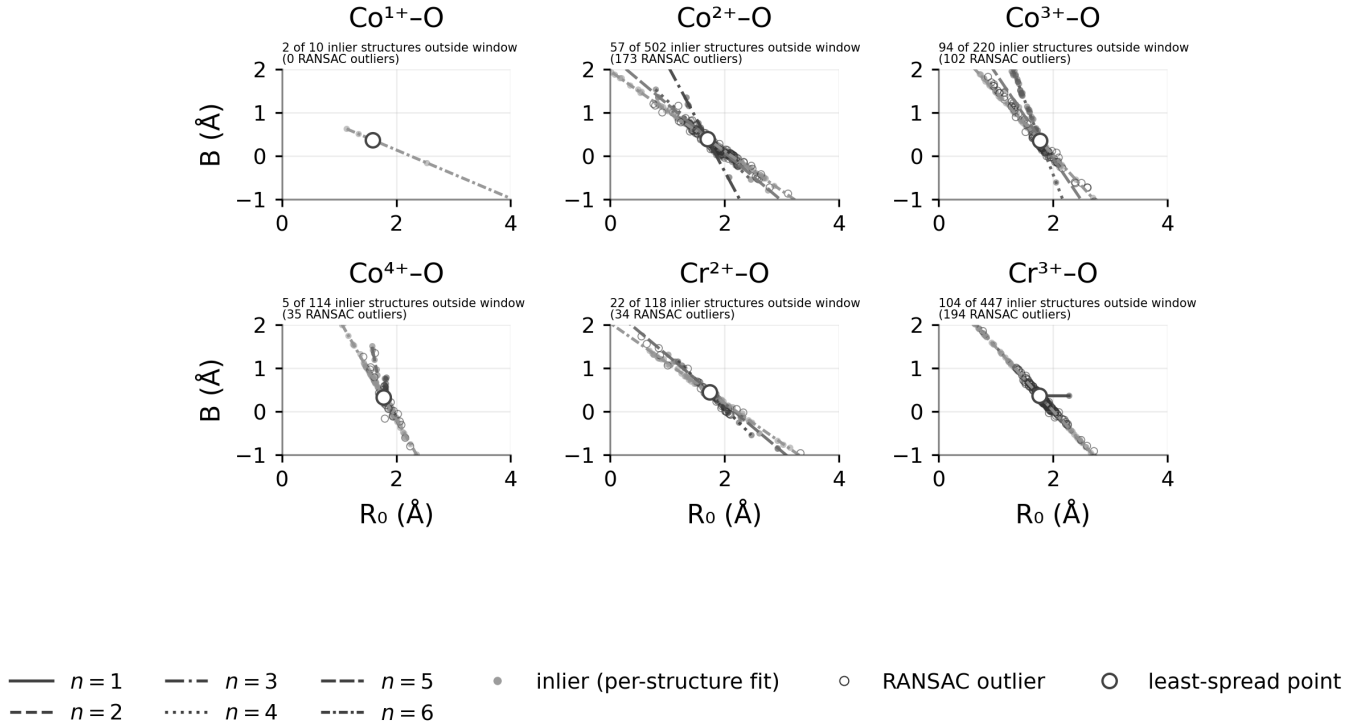
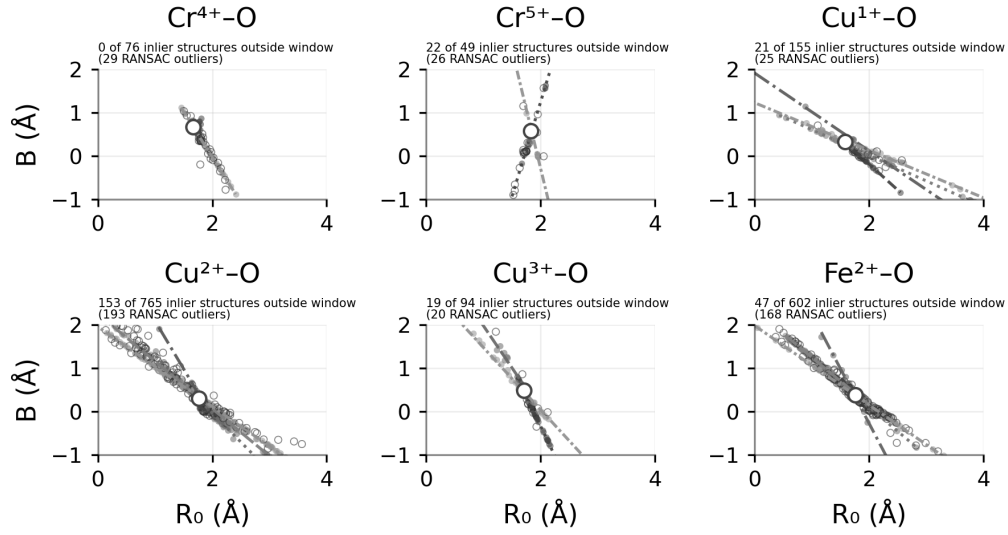
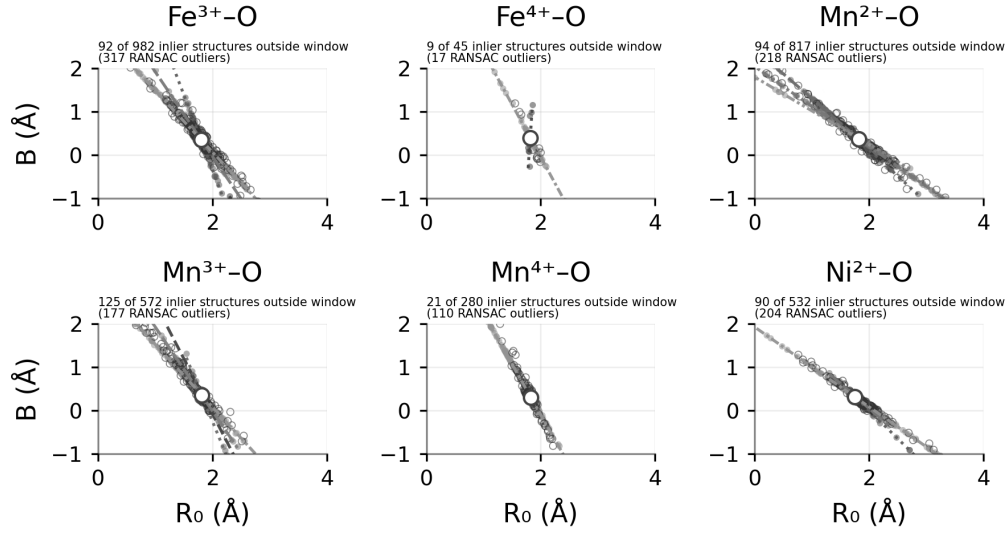


FIG. S4. n -resolved B vs. R_0 for oxidation-state-resolved 3d X-O pairs (Co^{1+} , Co^{2+} , Co^{3+} , Co^{4+} , Cr^{2+} , Cr^{3+}).



— $n=1$ - - - $n=3$ - - - $n=5$ • inlier (per-structure fit) ○ RANSAC outlier ○ least-spread point
 - - - $n=2$ ····· $n=4$ - - - $n=6$

FIG. S5. n -resolved B vs. R_0 for oxidation-state-resolved 3d X-O pairs (Cr⁴⁺, Cr⁵⁺, Cu¹⁺, Cu²⁺, Cu³⁺, Fe²⁺).



- - - $n=2$ ····· $n=4$ - - - $n=6$ - - - $n=8$ ○ RANSAC outlier ○ least-spread point
 - - - $n=3$ - - - $n=5$ ····· $n=7$ • inlier (per-structure fit)

FIG. S6. n -resolved B vs. R_0 for oxidation-state-resolved 3d X-O pairs (Fe³⁺, Fe⁴⁺, Mn²⁺, Mn³⁺, Mn⁴⁺, Ni²⁺).

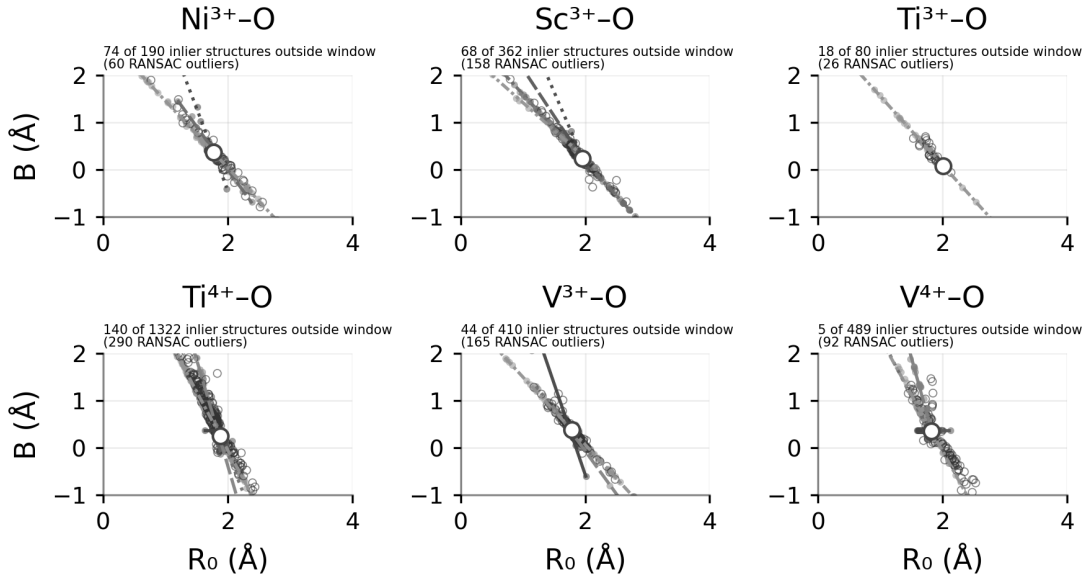


FIG. S7. n -resolved B vs. R_0 for oxidation-state-resolved 3d X-O pairs (Ni^{3+} , Sc^{3+} , Ti^{3+} , Ti^{4+} , V^{3+} , V^{4+}).

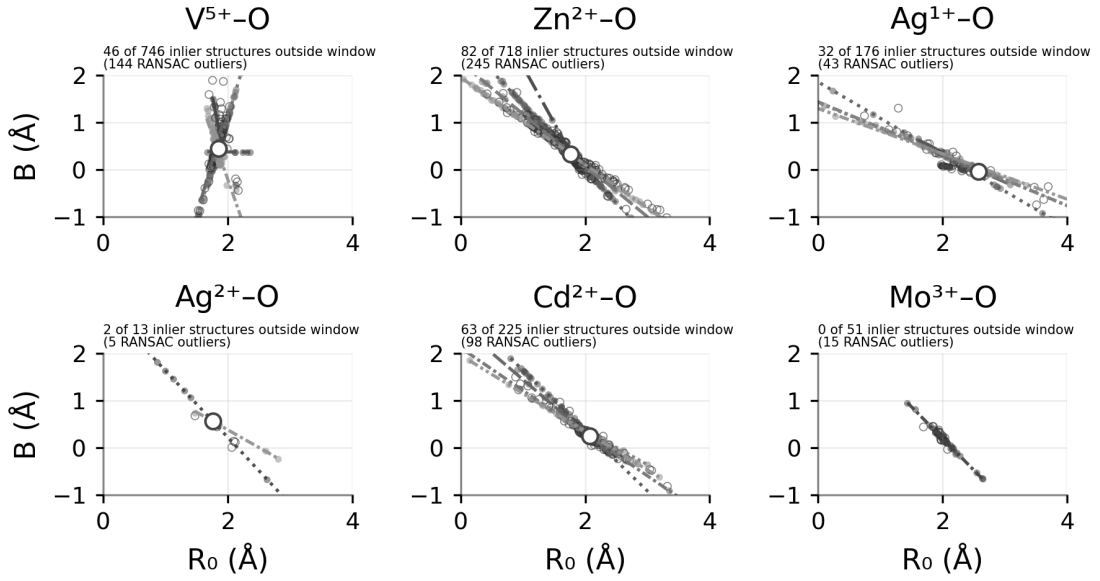


FIG. S8. n -resolved B vs. R_0 for oxidation-state-resolved 3d/4d X-O pairs (V^{5+} , Zn^{2+} , Ag^{1+} , Ag^{2+} , Cd^{2+} , Mo^{3+}).

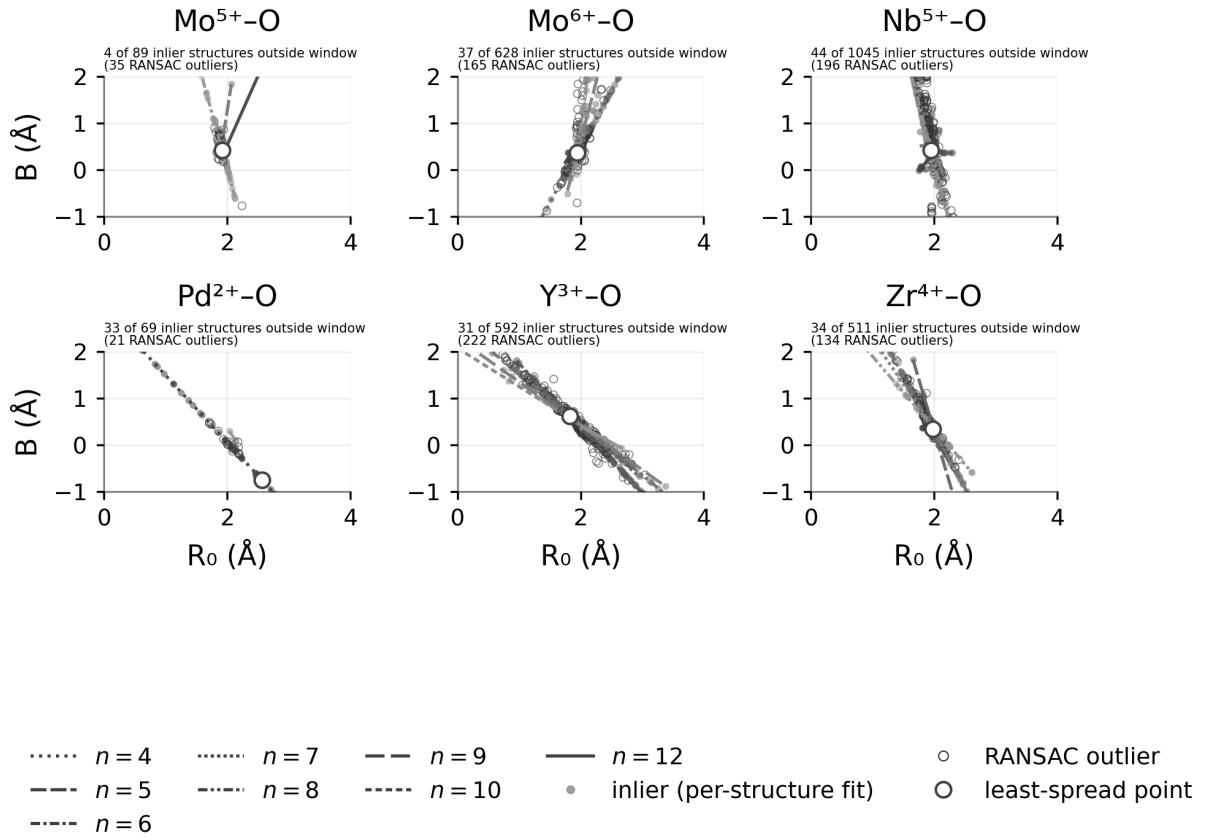


FIG. S9. n -resolved B vs. R_0 for oxidation-state-resolved 4d X-O pairs (Mo^{5+} , Mo^{6+} , Nb^{5+} , Pd^{2+} , Y^{3+} , Zr^{4+}).

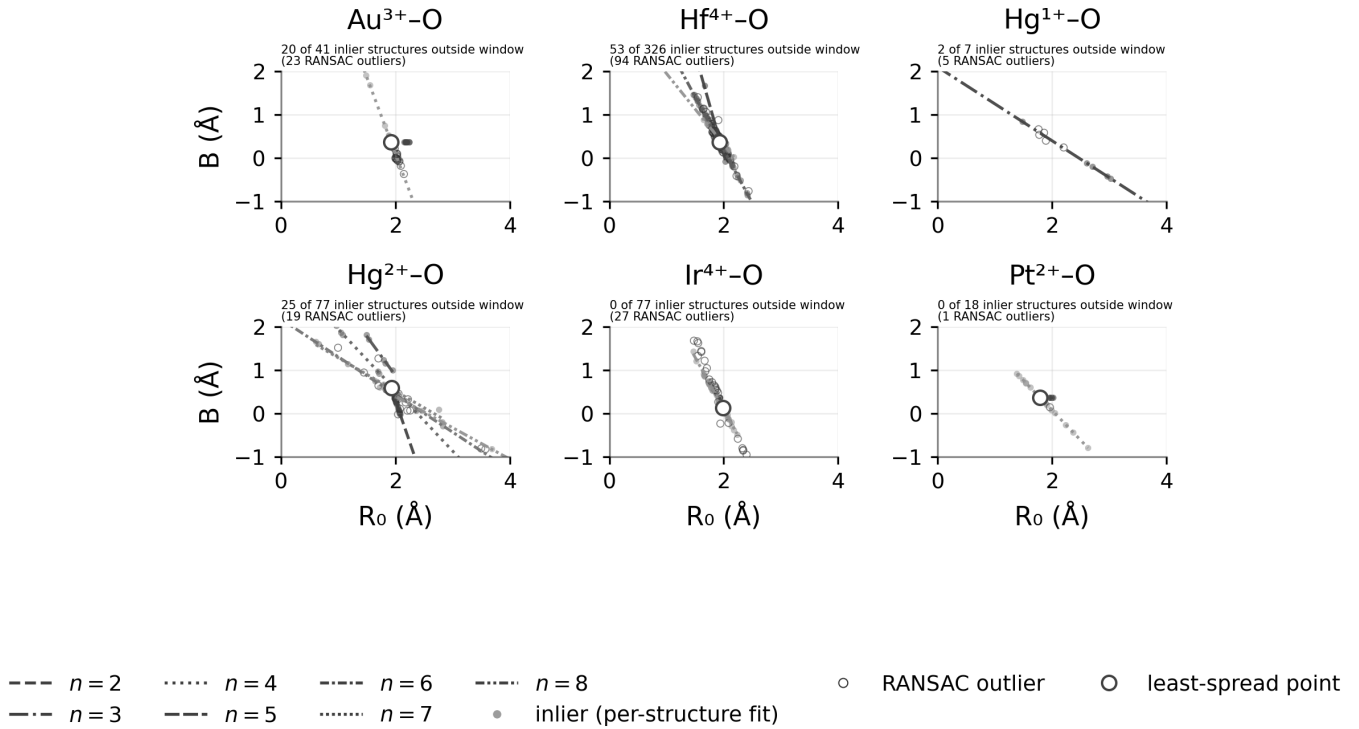


FIG. S10. n -resolved B vs. R_0 for oxidation-state-resolved 5d X-O pairs (Au^{3+} , Hf^{4+} , Hg^{1+} , Hg^{2+} , Ir^{4+} , Pt^{2+}).

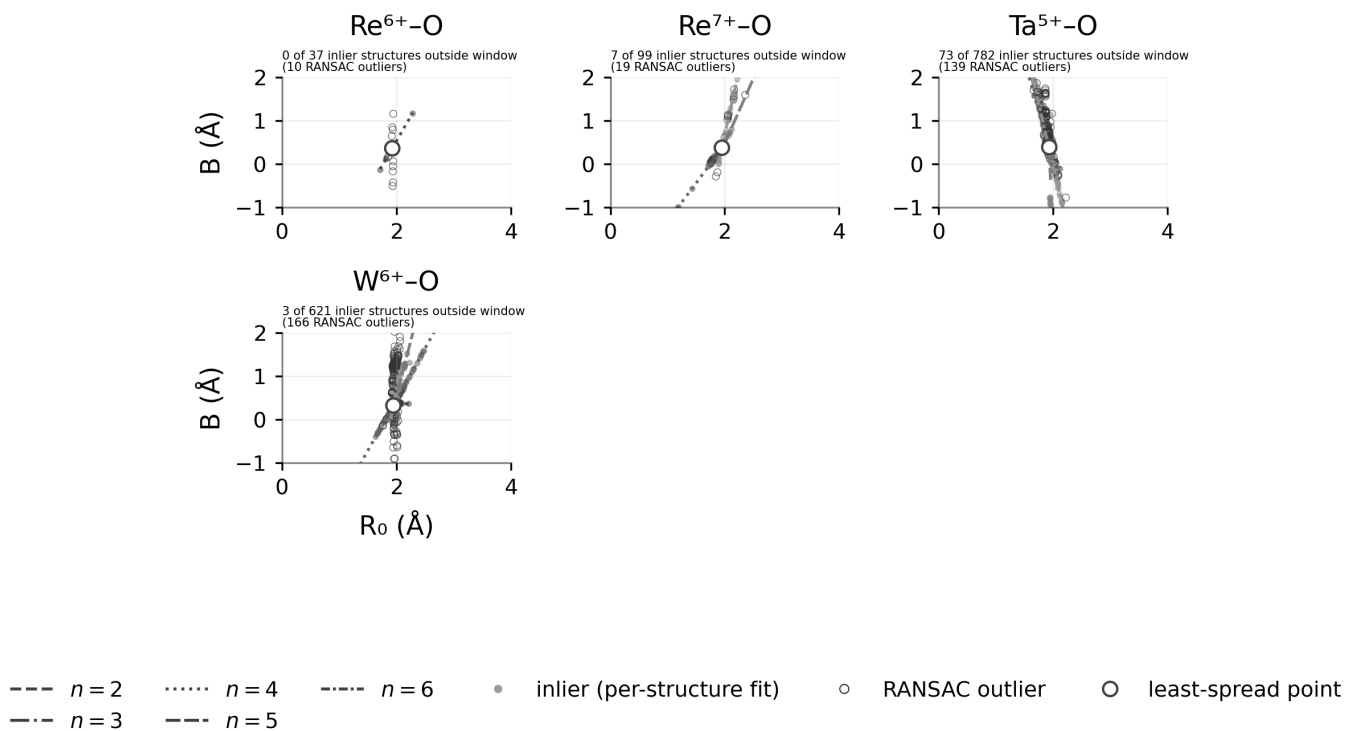


FIG. S11. n -resolved B vs. R_0 for oxidation-state-resolved 5d X-O pairs (Re^{6+} , Re^{7+} , Ta^{5+} , W^{6+}).

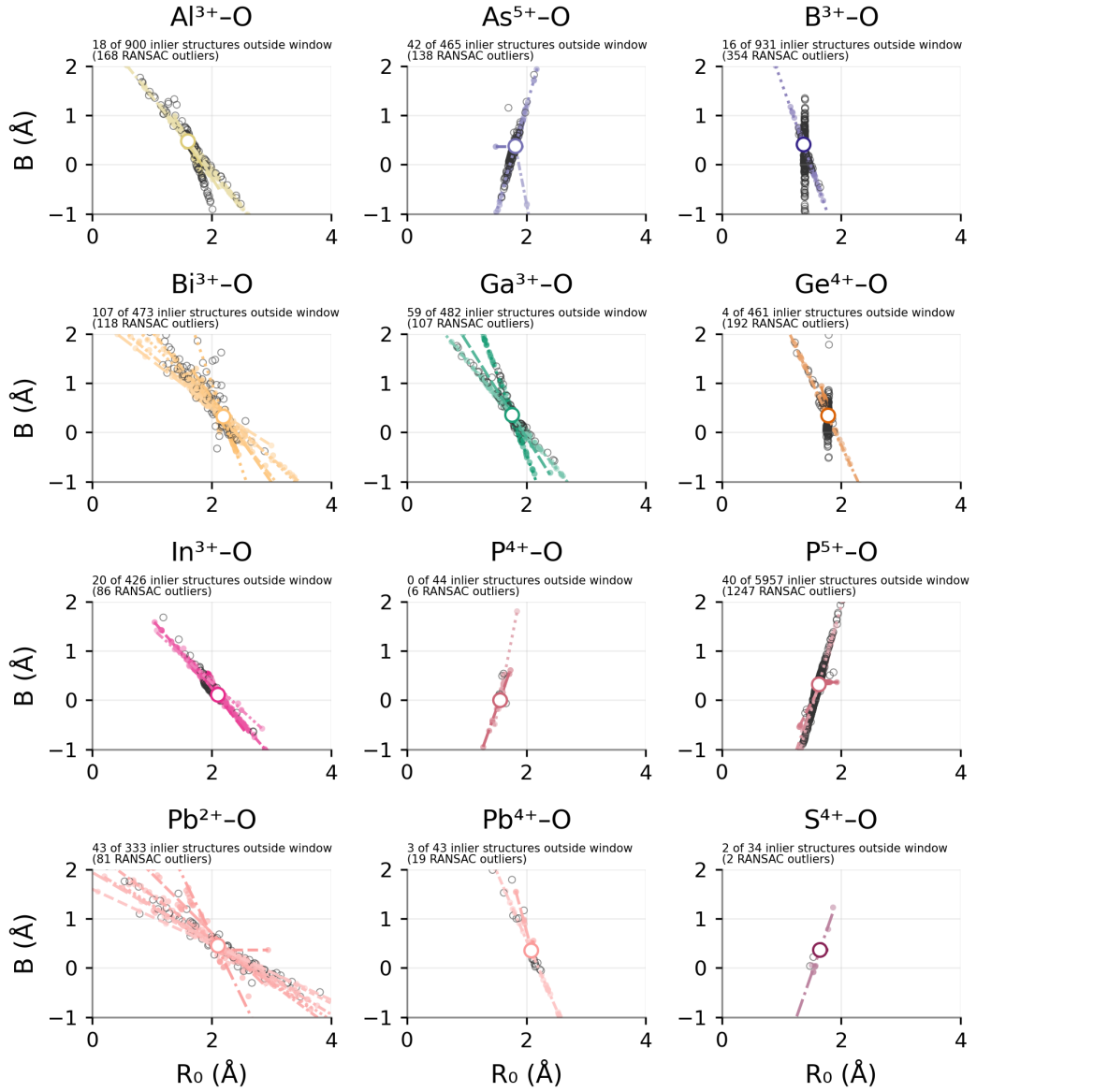


FIG. S12. n -resolved B vs. R_0 for oxidation-state-resolved p -block X-O pairs (Al³⁺-S⁴⁺).

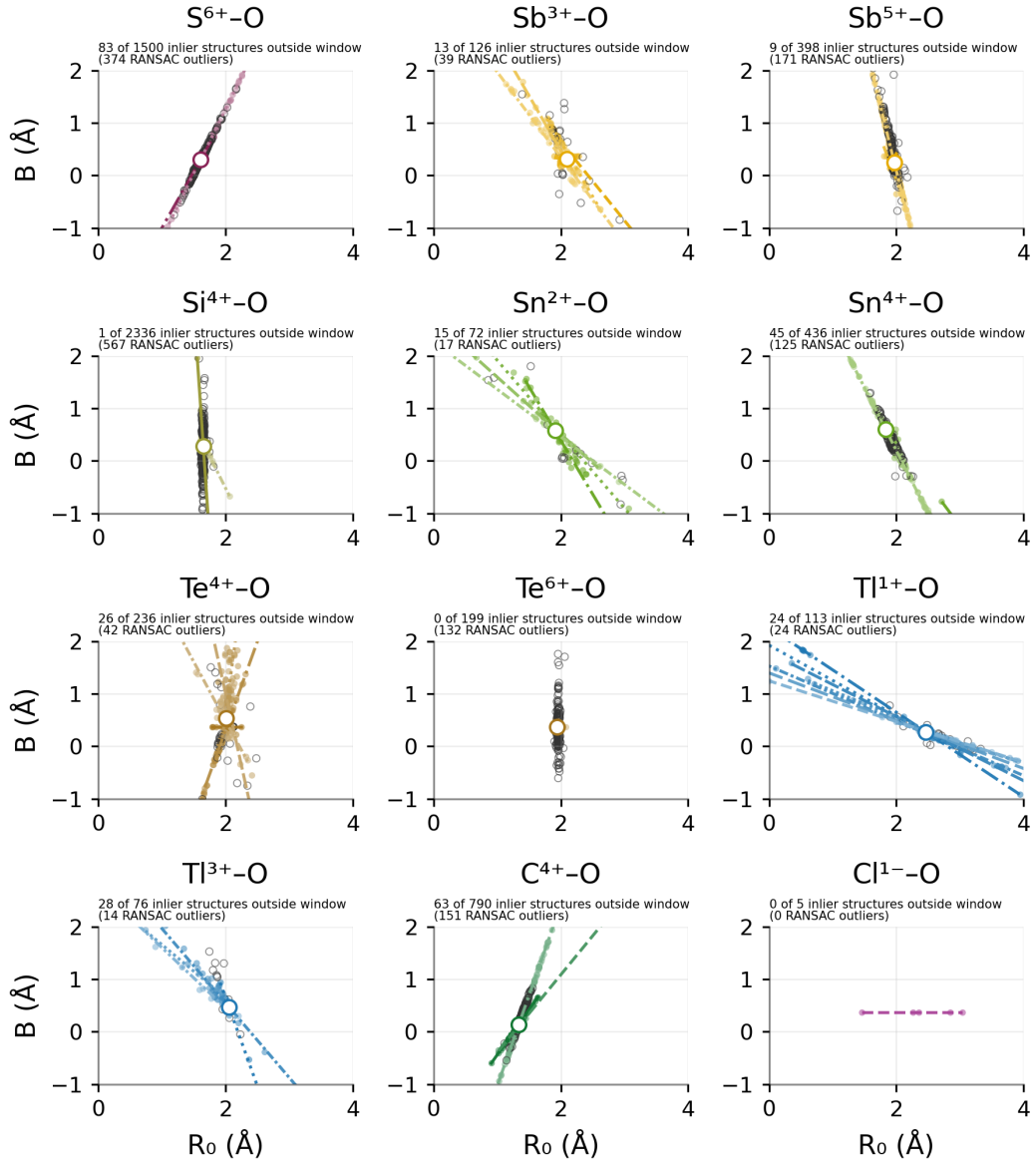


FIG. S13. n -resolved B vs. R_0 for oxidation-state-resolved p -block X-O pairs (S^{6+} , Sb^{3+} , Sb^{5+} , Si^{4+} , Sn^{2+} , Sn^{4+} , Te^{4+} , Te^{6+} , Tl^{1+} , Tl^{3+} , C^{4+} , Cl^{1-}).

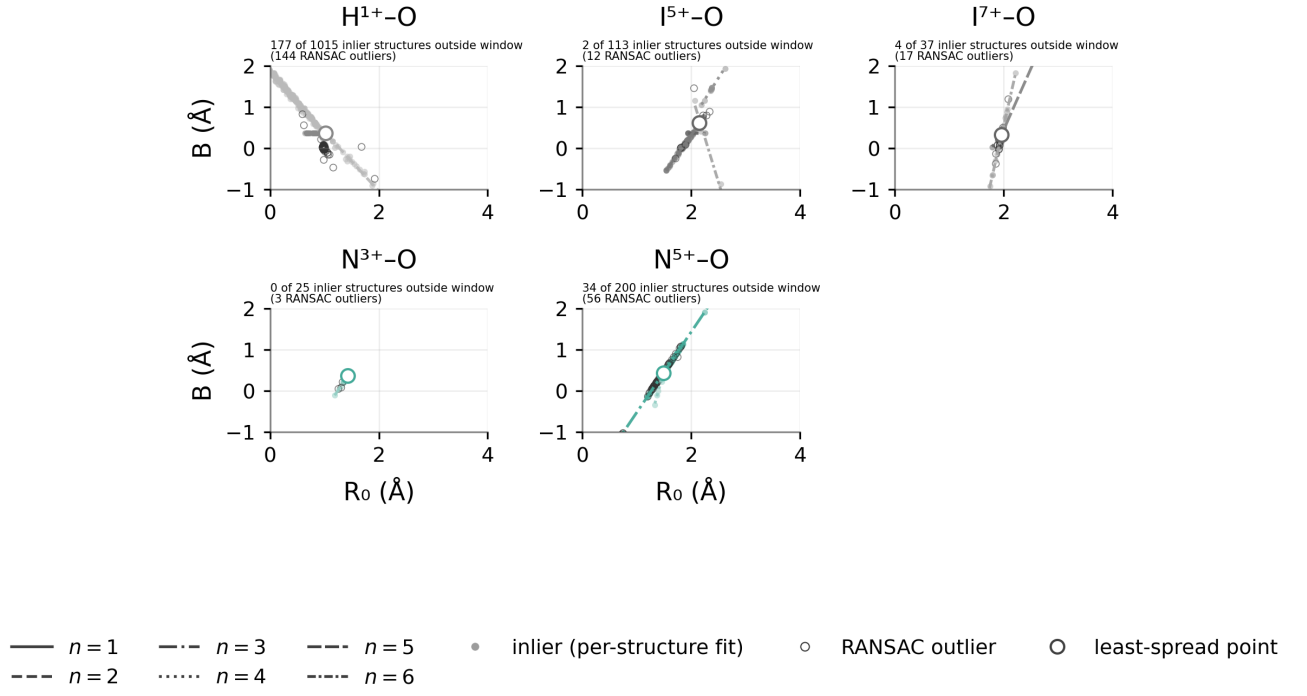


FIG. S14. n -resolved B vs. R_0 for the remaining p -block and hydrogen X-O pairs (H^{1+} , I^{5+} , I^{7+} , N^{3+} , N^{5+}).

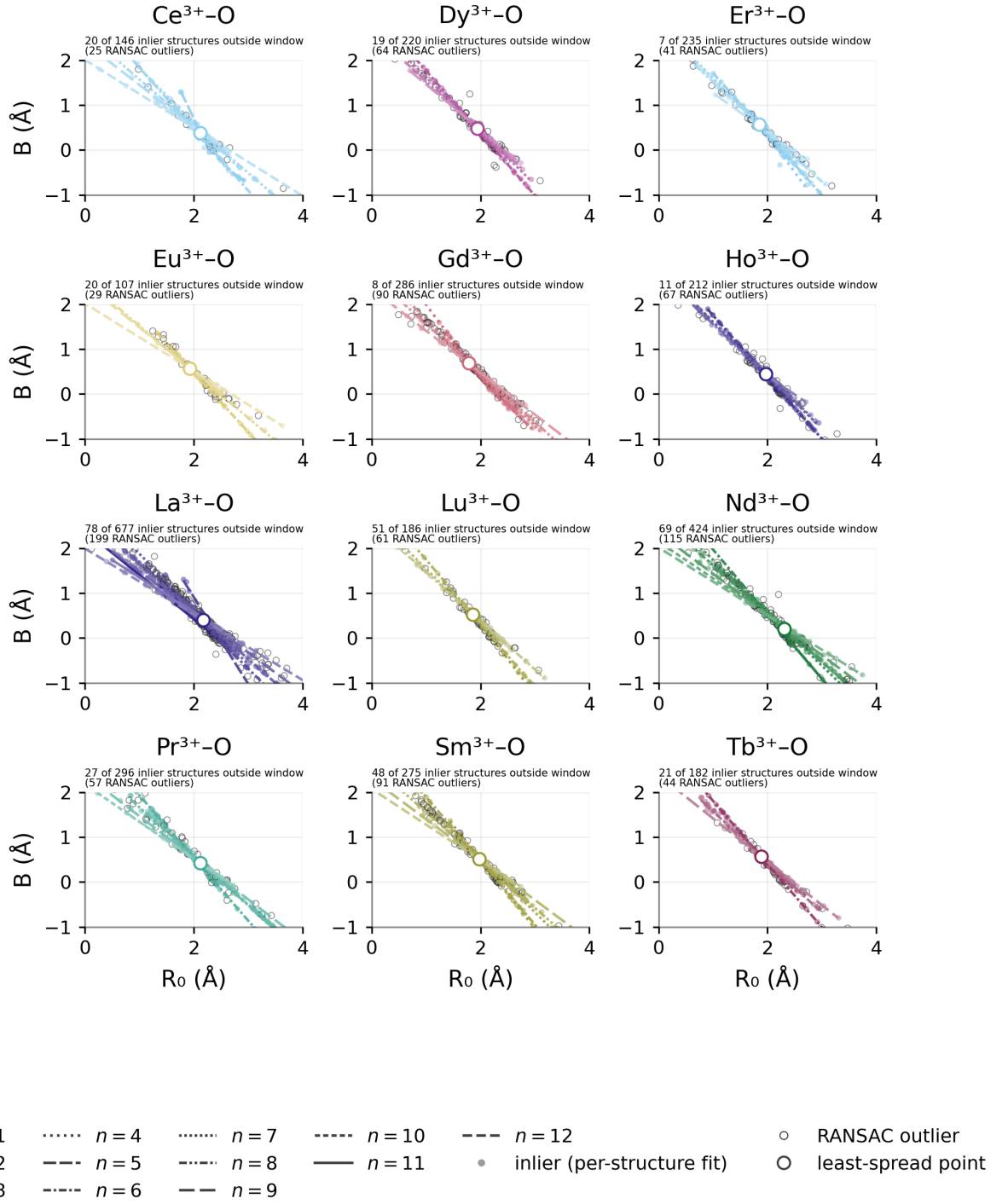


FIG. S15. n -resolved B vs. R_0 for f -block trivalent lanthanide X-O pairs (Ce³⁺–Tb³⁺).

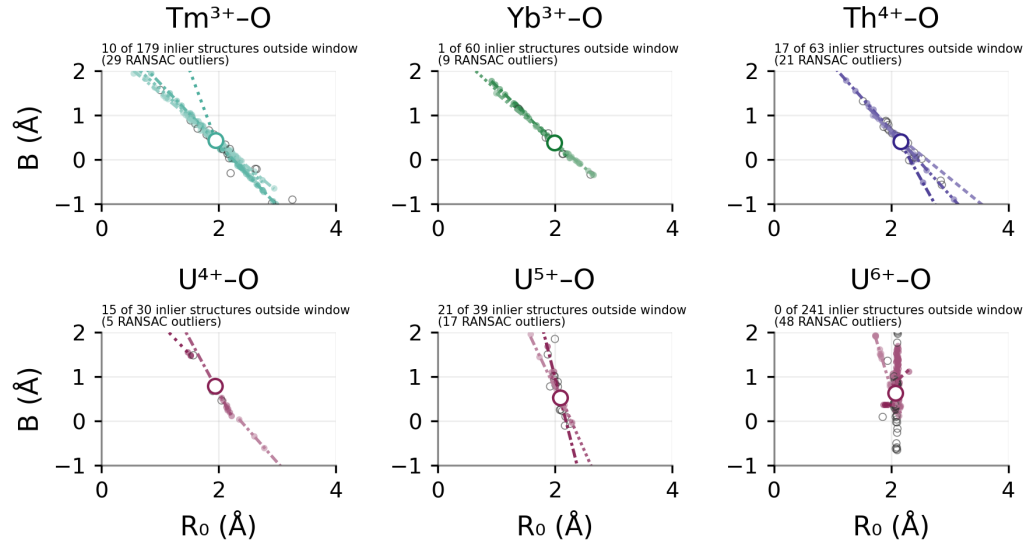


FIG. S16. n -resolved B vs. R_0 for the remaining f -block lanthanide and actinide X-O pairs (Tm³⁺-Yb³⁺ and Th⁴⁺-U⁶⁺).

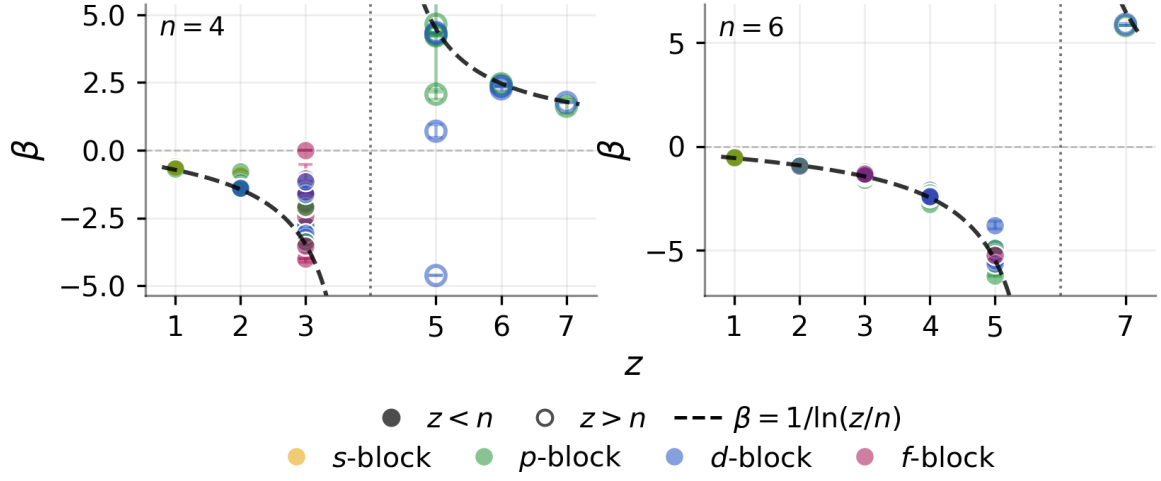


FIG. S17. Fitted slope β versus formal charge z for the individual cation–oxygen species summarized in main-text Fig. 1, at $n = 4$ (left) and $n = 6$ (right). Filled markers show species with $z < n$ (negative β), open markers species with $z > n$ (positive β), and dashed curves the parameter-free prediction $\beta = 1/\ln(z/n)$. Markers are colored by cation block and drawn semi-transparent, so darker points mark loci where the fitted slopes of multiple species coincide. Error bars show the standard error σ_β of each fitted slope. Most are smaller than the marker.

III. PER-SPECIES SLOPES AT FIXED COORDINATION

Figure S17 resolves the box-and-whisker summary of main-text Fig. 1 into the individual cation–oxygen species from which the coordination-number-resolved fits above were obtained.

IV. CHARACTERISTIC PAIRS AND BOND-LENGTH IDENTITY

Figure S18 shows the characteristic pairs (R_0^*, B^*) for Group 1 (including H^+), Group 2, and the trivalent lanthanides.

For each species and coordination shell n , the per-structure (R_0, B) pairs define a line $B = \beta^{(n)} R_0 + \beta_0^{(n)}$, fitted by the robust procedure described above. The characteristic pair is the weighted least-spread intersection of those n -resolved lines. All available coordination lines enter, with no R^2 screening at this stage. If the final line fit for shell n retains N_n inlier structures, we assign weight $w_n = N_n$ and define the weighted means

$$\bar{\beta} = \frac{\sum_n w_n \beta^{(n)}}{\sum_n w_n}, \quad \bar{\beta}_0 = \frac{\sum_n w_n \beta_0^{(n)}}{\sum_n w_n}.$$

The characteristic abscissa is then

$$R_0^* = - \frac{\sum_n w_n (\beta^{(n)} - \bar{\beta}) (\beta_0^{(n)} - \bar{\beta}_0)}{\sum_n w_n (\beta^{(n)} - \bar{\beta})^2},$$

which is the point that minimizes the weighted variance of the line-by-line values $B_n(R_0) = \beta^{(n)} R_0 + \beta_0^{(n)}$. The characteristic ordinate and its spread are

$$B^* = \frac{\sum_n w_n B_n(R_0^*)}{\sum_n w_n}, \quad \sigma_B = \left[\frac{\sum_n w_n (B_n(R_0^*) - B^*)^2}{\sum_n w_n} \right]^{1/2}.$$

The tabulated n_{CN} counts how many distinct coordination shells contribute to that intersection, while the weights entering the intersection are the fitted structure counts in those shells.

Near-coincidence of same-branch lines. For many species the n -resolved lines appear nearly coincident. This is expected from the pole form. Subtracting two predicted slopes gives

$$\beta^{(n_2)} - \beta^{(n_1)} = \frac{\ln(n_2/n_1)}{\ln(z/n_1) \ln(z/n_2)}.$$

The numerator involves only the coordination ratio and is small for neighboring shells. When z lies on the same side of both coordination numbers and well away from each, the denominator is of order unity, so same-branch lines are nearly parallel over the plotted windows and their pairwise crossing is delicate. As z approaches either coordination number the corresponding line steepens toward the pole, the slope difference grows, and the crossing sharpens. When instead $n_1 < z < n_2$ the two slopes have opposite signs and the crossing is sharp. Equation (7) of the main text makes the conditioning explicit. The crossing ordinate is the shell dilation $[\bar{R}^{(n_2)} - \bar{R}^{(n_1)}]/\ln(n_2/n_1)$, so the intersection is well constrained when the corpus resolves the dilation of the shell with coordination number and only weakly constrained when it does not. The characteristic pair is therefore never computed from a single pair of lines. The weighted least-spread construction above pools every available coordination line, weights each by its fitted structure count, and reports the spread σ_B at R_0^* as the conditioning diagnostic in Table S1. Across the 85 species with three or more lines the median σ_B/B^* is 5.8%, and 62 fall below 10%. The weakest case is Ag^{1+} , whose near-zero $B^* = 0.011 \text{ \AA}$ inflates the relative spread. The 18 species with only two lines have $\sigma_B = 0$ by construction and are marked provisional.

Figure S19 validates the bond-length identity $R_0 \approx \bar{R} + B \ln(z/n)$ across 44,844 screening-branch cation-oxygen structures (fitted single-valence records with a defined mean bond length, $B > 0$, $0 < R_0 < 4 \text{ \AA}$, and $z \neq n$), with Pearson $r = 0.976$ and MAE = 0.019 $\text{\AA}$.

V. COMPARISON WITH PUBLISHED SOFTNESS PARAMETERS

The characteristic softness B^* of Eq. (7) of the main text is a single species-level constant obtained by aggregating per-structure fits across the corpus, and is therefore directly comparable to published bond-valence parameter sets, each of which assigns one softness per cation-anion pair by pooling all structures containing that pair into a single fit. Brown and Altermatt fix $B = 0.37 \text{ \AA}$ for every pair and fit only R_0 [3]. The softBV parameterization fits one b per pair from pooled structure data with coordination-dependent cutoffs [4, 5], and Gagné and Hawthorne refit both parameters per pair from a large pooled corpus [6]. Species are matched to the published sets by element and oxidation state.

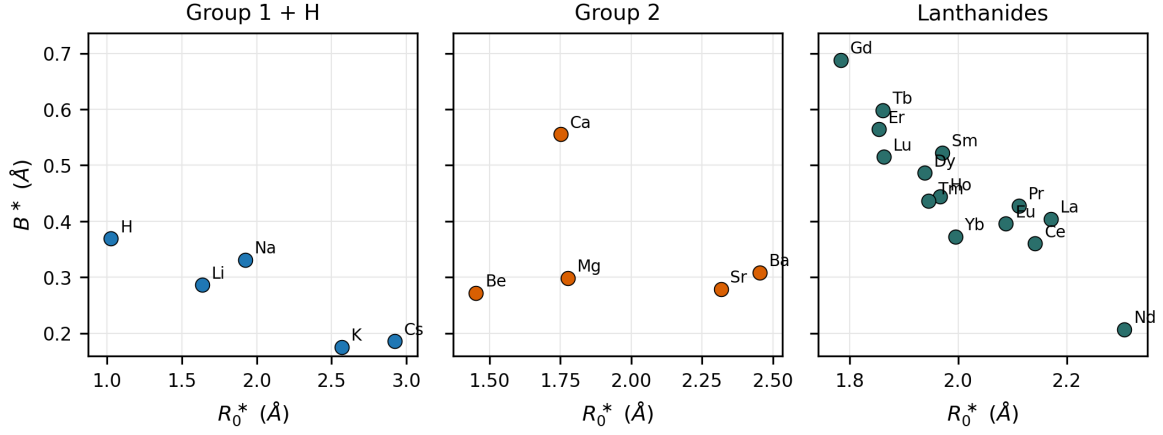


FIG. S18. Characteristic (R_0^*, B^*) for Group 1 including H^+ (left), Group 2 (center), and trivalent lanthanides (right).

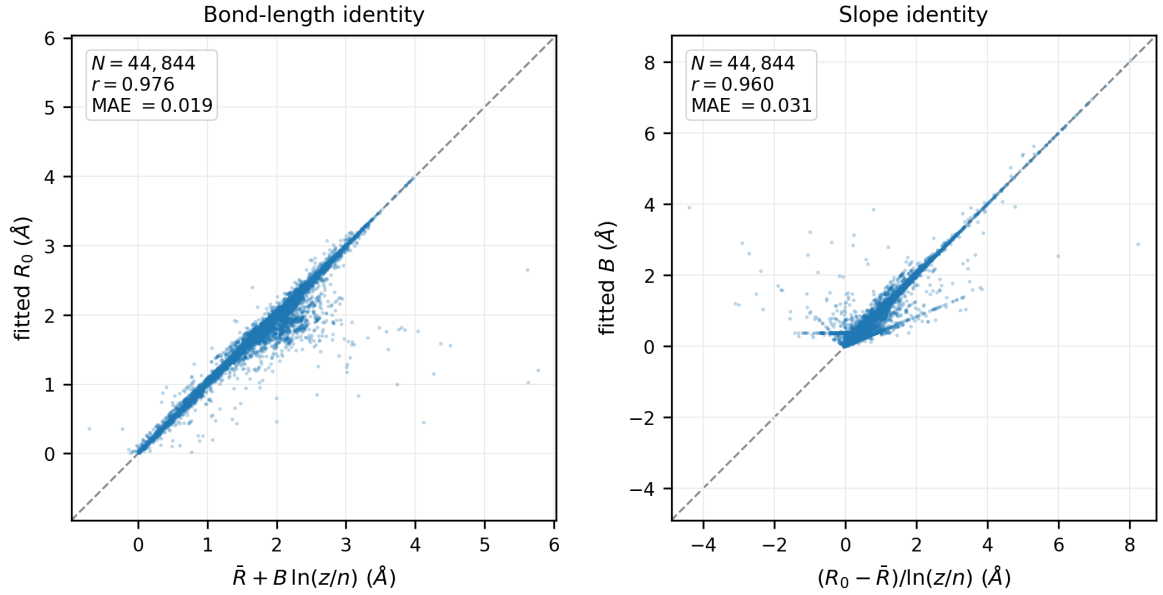


FIG. S19. Left: parity between fitted R_0 and $\bar{R} + B \ln(z/n)$ for 44,844 screening-branch cation–oxygen structures. Right: parity between fitted B and $(R_0 - \bar{R})/\ln(z/n)$ for the same set.

Distribution-level agreement. Across the 101 screening-branch species, B^* has mean 0.38 ± 0.15 Å, median 0.37 Å, interquartile range 0.31–0.44 Å, and full range 0.004–0.918 Å, clustering near the conventional assignment $B = 0.37$ Å [3]. The softBV set spans 0.34–0.62 Å over 155 cation–oxygen pairs (mean 0.45 ± 0.05 Å), and the Gagné–Hawthorne set spans 0.25–0.66 Å over 135 pairs (mean 0.40 ± 0.06 Å). On matched species the mean offset of B^* from the Gagné–Hawthorne values is -0.01 Å (92 species, mean absolute difference 0.11 Å) and from softBV -0.06 Å (99 species, mean absolute difference 0.13 Å). The distribution is therefore consistent with the spread determined by flexible per-pair fitting [6] while sitting lower than the values deployed by softBV [4, 5]. The extreme B^* tails are the weakly conditioned intersections flagged in Table S1. Restricting to the well-conditioned species, with at least three coordination lines and $\sigma_B/B^* \leq 20\%$, leaves 78 species with mean 0.39 ± 0.12 Å, median 0.37 Å, and range 0.14–0.79 Å.

Species-level decorrelation. Species by species, B^* is uncorrelated with either published set ($r = 0.09$ against softBV and $r = -0.01$ against Gagné–Hawthorne, essentially unchanged on the well-conditioned subset). This is not specific to the present method. The two published sets correlate only weakly with each other ($r = 0.26$, mean absolute difference 0.065 Å over 130 common pairs), because a pooled per-pair fit selects one point in the shallow (R_0, B) valley documented by Gagné and Hawthorne [6], and where that point falls depends on which coordination numbers and bonding environments dominate the training corpus. The per-material fits used here resolve the valley instead of averaging over it. At fixed species and coordination number the per-structure (R_0, B) pairs fall on the lines

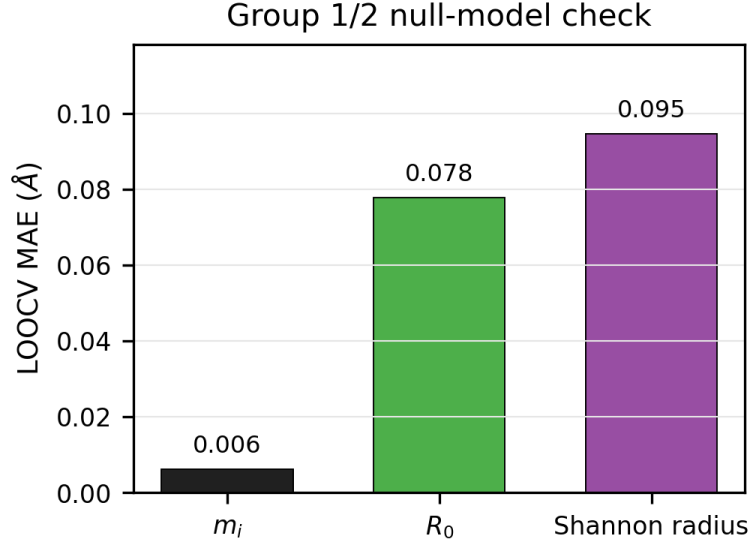


FIG. S20. Leave-one-out null-model check for the ten Group 1/Group 2 binary oxides used in Fig. 2 of the main text. Bars report the mean absolute error of linear regressions against the bond-valence shell centroid m_i , the fitted intercept R_0 , and the Shannon cation radius.

of slope $1/\ln(z/n)$ shown in the figures above, so the covariance that pooled fitting suppresses carries the physical content, and B^* is defined geometrically as the least-spread intersection of those lines rather than as a corpus average. The two estimands therefore agree in scale and center but need not agree pair by pair.

VI. *s*-BLOCK SCREENING-CENTROID CONTROLS

For the ten Group 1/Group 2 ground-state binary oxides used in Fig. 2 of the main text, the DFT screening centroid r_{eff} is compared against three linear descriptors: the bond-valence shell centroid m_i , the fitted bond-valence intercept R_0 , and the Shannon cation radius. Leave-one-out mean absolute errors are 0.006 Å for m_i , 0.078 Å for R_0 , and 0.095 Å for the Shannon-radius control, showing that the $r_{\text{eff}}-m_i$ collapse is substantially sharper than a generic monotone size trend.

Each oxide contributes one point (m_i, r_{eff}) to main-text Fig. 2. For a given oxide, the bond-valence shell centroid is computed from the same fitted (R_0, B) and coordination number n used for its bond-valence record, taking the n shortest M–O distances in the ground-state structure and forming

$$p_{ij} = \frac{e^{-R_{ij}/B}}{\sum_{k=1}^n e^{-R_{ik}/B}}, \quad m_i = \sum_{j=1}^n p_{ij} R_{ij}.$$

The DFT-side centroid is then built bond by bond over all oxygen neighbors with $R_{ij} \leq 3.5$ Å. Each such bond contributes one oxygen-side path centroid $r_{\text{eff}}^{(\text{path})}$, and the oxide-level r_{eff} plotted in main-text Fig. 2 is the simple arithmetic mean of those finite bond-path centroids.

The offset in the main-text regression is quantified directly from that same discrete bond-path construction. Defining the oxygen-side pre-cusp valence centroid shift by

$$\Delta_{\text{O, val}} := \langle R_{ij} - r_{\text{eff}}^{(\text{path})} \rangle,$$

let $\{r_g, \rho_g\}_{g=1}^{200}$ denote the total-density samples on the nearest-image straight M–O segment at 200 uniformly spaced points. The bond critical-point proxy is the interior density minimum on the interval $0.15R_{ij} < r < 0.85R_{ij}$. Denote its location and value by r_{BCP} and ρ_{BCP} . The oxygen-side pre-cusp window is the subset of sampled points with $r_{\text{BCP}} \leq r_g \leq 0.93R_{ij}$, finite ρ_g , and positive finite Thomas–Fermi screening length $\lambda_{\text{TF}}(\rho_g)$. The Thomas–Fermi length enters here only as a positivity filter on the sample window and does not appear in the centroid weights or in any of the regression results. The `thomas_fermi` tag in the main-text Fig. 2 file label reflects only the screening interpretation of r_{eff} , not its construction. On that discrete window we use the excess-density weights $w_g = \max[\rho_g - \rho_{\text{BCP}}, 0]$. In

the rare case that these clipped weights vanish identically, we use $w_g = \rho_g$ instead. Thus

$$r_{\text{eff}}^{(\text{path})} = \frac{\sum_g w_g r_g}{\sum_g w_g},$$

and the oxide-level centroid used in main-text Fig. 2 is the arithmetic mean of $r_{\text{eff}}^{(\text{path})}$ over those first-shell M–O bonds. No spline fitting, path relaxation, or topological bond-path search is introduced beyond this fixed straight-segment sampling rule. With that definition, the ten-oxide s -block shift fit is $\Delta_{\text{O, val}} = 0.0613 m_i + 0.2394 \text{ \AA}$ ($R^2 = 0.963$), with mean $0.3910 \pm 0.0269 \text{ \AA}$. Equivalently, $r_{\text{eff}} = 0.9387 m_i - 0.2394 \text{ \AA}$, whose rounded form is the main-text relation. The intercept is therefore the zero-radius limit of a directly measured oxygen-side screening shift, not an unconstrained empirical parameter.

Species-level screening length from the characteristic pair. The shell-level Yukawa inversion in the main text is written in terms of the representative radius R' , chosen operationally as the Gibbs shell centroid m_i . To pass from that shell-level relation to a species-level constant, we evaluate the inversion at the characteristic fixed point (R_0^*, B^*) on the characteristic reference shell defined by $m_i = R_0^*$. Using the exact identity

$$R_0 = m_i + B(\ln q_i - H_i),$$

this reference-shell condition is equivalent at the fixed point to $H_i = \ln q_i$, so the representative radius entering the Yukawa inversion is $R' = R_0^*$. Substituting $B = B^*$ and $R' = R_0^*$ into the positive screening branch therefore gives

$$\lambda^* = \frac{-R_0^* + \sqrt{R_0^{*2} + 4B^*R_0^*}}{2},$$

which is the species-level screening length reported in Table S1. For the two anti-screening species ($B^* < 0$) the same branch-continuous root is real whenever $B^* \geq -R_0^*/4$, a condition both satisfy. Their negative λ^* values are tabulated as signed Yukawa-branch diagnostics rather than physical screening lengths. The second root of the same quadratic is tabulated as λ_-^* for every species. On the anti-screening branch it also lies in the admissible interval $(-R_0^*, 0)$. On the screening branch it lies below $-R_0^*$, outside the admissible interval, and is reported for completeness.

Crystal-chemical anchoring of the second root. Because the two roots of the inversion quadratic sum to $-R_0^*$, the magnitude of the second root on the screening branch is $|\lambda_-^*| = R_0^* + \lambda^*$, the characteristic shell radius extended by one screening length. Figure S21 compares this quantity with an independent crystal-chemical scale. Each species is assigned a population-weighted Shannon crystal radius, obtained by interpolating the Shannon crystal radii [7] linearly in coordination number and averaging over the coordination numbers observed for that species in the fitted corpus, weighted by fitted structure counts. The high-spin radius is used where both spin states are tabulated. Unit-slope fits $|\lambda_-^*| = r_{\text{cr}} + c(z)$ describe each formal-charge class with 0.05–0.13 $\text{\AA}$ rms scatter, and the fitted offsets follow $c(z) = 1.05 + 0.092z \text{ \AA}$ ($r = 0.97$, inset). The constant term is an oxygen-radius-scale length, and the increment raises the effective cation–oxygen contact distance by about 0.09 $\text{\AA}$ per unit of formal charge. The figure shows the 96 cationic screening-branch species with tabulated Shannon radii. H^+ , P^{4+} , Co^+ , and Sn^{2+} lack entries at any coordination number, and Cl^- and the two anti-screening species are excluded.

VII. SCREENING LENGTHS FOR THE ANALYZED SPECIES

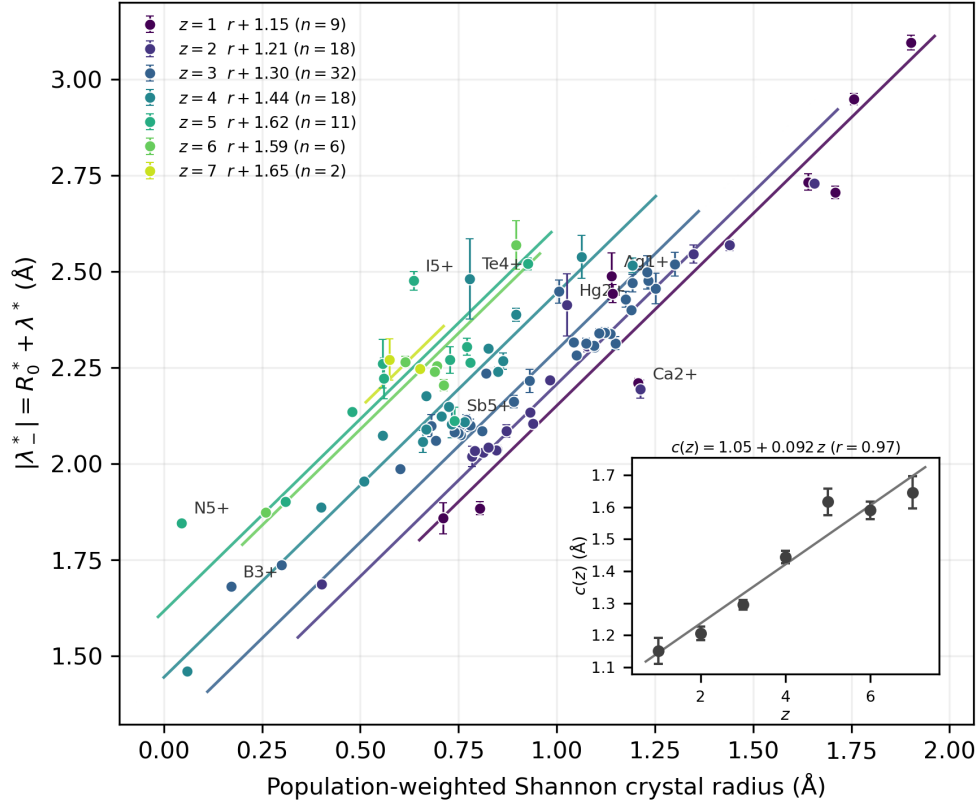


FIG. S21. Magnitude of the second inversion root, $|\lambda^*| = R_0^* + \lambda^*$, versus the population-weighted Shannon crystal radius [7] for 96 screening-branch cation species. Lines are unit-slope fits $|\lambda^*| = r_{cr} + c(z)$ for each formal charge z . Error bars are the propagated σ_λ of Table S1. The inset shows the fitted offsets $c(z)$ with standard errors and the linear law $c(z) = 1.05 + 0.092z$ ($r = 0.97$). Labels mark the largest per-class residuals.

TABLE S1: Characteristic pairs (R_0^*, B^*) for all 103 cation-oxygen species and Yukawa screening lengths λ^* , ordered by atomic number. All values are in Å. λ^* is computed from the branch-continuous inversion $\lambda^* = (-R_0^* + \sqrt{R_0^{*2} + 4B^*R_0^*})/2$ in the main text, which is real for $B^* \geq -R_0^*/4$. For the 101 screening-branch species ($B^* > 0$) it is positive and physical. For the two anti-screening species (Mo^{3+} and Sb^{3+} , $B^* < 0$) the same root is negative and is tabulated as a signed Yukawa-branch diagnostic rather than a physical screening length, because the first-order expansion does not adequately describe their bonding. The second root of the same quadratic, $\lambda_-^* = (-R_0^* - \sqrt{R_0^{*2} + 4B^*R_0^*})/2$, is listed for every species in the λ_-^* column. On the anti-screening branch it also lies in the admissible interval $(-R_0^*, 0)$. On the screening branch it lies below $-R_0^*$, outside the admissible interval, and is reported for completeness. n_{CN} is the number of distinct coordination shells used to determine (R_0^*, B^*) . Parenthetical uncertainties on B^* are the weighted standard deviation σ_B of the n -resolved lines at R_0^* . Uncertainties on λ^* are propagated via $\sigma_\lambda = |\partial\lambda^*/\partial B^*|\sigma_B$. Entries with $n_{\text{CN}} = 2$ have $\sigma_B = 0$ by construction and should be regarded as provisional.

Species	Block	R_0^*	B^*	λ^*	λ_-^*	n_{CN}
H^{1+}	s	1.026	0.369(9)	0.288(6)	-1.314(6)	9
Li^{1+}	s	1.636	0.286(22)	0.248(17)	-1.884(17)	10
Be^{2+}	s	1.452	0.272(0)	0.234(0)	-1.686(0)	2
B^{3+}	p	1.376	0.372(9)	0.304(6)	-1.681(6)	7
C^{4+}	p	1.337	0.135(4)	0.123(3)	-1.460(3)	3

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TABLE S1 – continued from previous page

Species	Block	R_0^*	B^*	λ^*	λ_-^*	n_{CN}
N^{3+}	p	1.431	0.370(0)	0.305(0)	-1.736(0)	2
N^{5+}	p	1.492	0.437(0)	0.354(0)	-1.845(0)	2
Na^{1+}	s	1.923	0.331(15)	0.288(11)	-2.211(11)	10
Mg^{2+}	s	1.776	0.298(16)	0.260(12)	-2.036(12)	5
Al^{3+}	p	1.648	0.407(9)	0.338(6)	-1.985(6)	6
Si^{4+}	p	1.652	0.269(9)	0.235(7)	-1.887(7)	5
P^{4+}	p	1.552	0.004(0)	0.004(0)	-1.556(0)	2
P^{5+}	p	1.625	0.324(7)	0.277(6)	-1.902(6)	4
S^{4+}	p	1.644	0.370(0)	0.311(0)	-1.955(0)	3
S^{6+}	p	1.610	0.307(0)	0.264(0)	-1.874(0)	2
Cl^{1-}	p	1.804	0.370(0)	0.315(0)	-2.119(0)	2
K^{1+}	s	2.568	0.175(24)	0.165(21)	-2.733(21)	12
Ca^{2+}	s	1.751	0.555(35)	0.443(24)	-2.195(24)	8
Sc^{3+}	d	1.961	0.222(19)	0.201(16)	-2.162(16)	6
Ti^{3+}	d	2.009	0.080(0)	0.077(0)	-2.086(0)	2
Ti^{4+}	d	1.881	0.248(43)	0.222(35)	-2.103(35)	6
V^{3+}	d	1.782	0.375(22)	0.318(16)	-2.100(16)	6
V^{4+}	d	1.821	0.353(15)	0.303(11)	-2.124(11)	5
V^{5+}	d	1.849	0.448(75)	0.373(53)	-2.221(53)	6
Cr^{2+}	d	1.744	0.435(14)	0.360(10)	-2.104(10)	3
Cr^{3+}	d	1.761	0.370(0)	0.314(0)	-2.075(0)	3
Cr^{4+}	d	1.665	0.670(0)	0.512(0)	-2.178(0)	2
Cr^{5+}	d	1.821	0.544(93)	0.439(63)	-2.260(63)	3
Mn^{2+}	d	1.830	0.354(6)	0.304(5)	-2.134(5)	7
Mn^{3+}	d	1.810	0.357(18)	0.305(13)	-2.116(13)	7
Mn^{4+}	d	1.830	0.296(0)	0.260(0)	-2.090(0)	2
Fe^{2+}	d	1.760	0.385(22)	0.325(16)	-2.085(16)	6
Fe^{3+}	d	1.799	0.352(9)	0.302(6)	-2.101(6)	5
Fe^{4+}	d	1.816	0.394(0)	0.333(0)	-2.149(0)	2
Co^{1+}	d	1.584	0.370(0)	0.310(0)	-1.893(0)	2
Co^{2+}	d	1.712	0.384(6)	0.323(5)	-2.035(5)	4
Co^{3+}	d	1.776	0.354(5)	0.303(3)	-2.078(3)	4
Co^{4+}	d	1.774	0.330(38)	0.284(29)	-2.058(29)	3
Ni^{2+}	d	1.766	0.304(8)	0.264(6)	-2.030(6)	5
Ni^{3+}	d	1.773	0.364(12)	0.310(9)	-2.083(9)	3
Cu^{1+}	d	1.585	0.319(53)	0.273(40)	-1.858(40)	4
Cu^{2+}	d	1.715	0.358(36)	0.304(27)	-2.019(27)	7
Cu^{3+}	d	1.708	0.481(42)	0.391(29)	-2.099(29)	3
Zn^{2+}	d	1.760	0.328(7)	0.283(5)	-2.043(5)	4
Ga^{3+}	p	1.756	0.357(5)	0.304(4)	-2.060(4)	4
Ge^{4+}	p	1.771	0.354(15)	0.302(11)	-2.073(11)	4
As^{5+}	p	1.812	0.382(1)	0.324(1)	-2.136(1)	3
Rb^{1+}	s	2.809	0.146(16)	0.140(15)	-2.949(15)	10
Sr^{2+}	s	2.317	0.279(17)	0.252(14)	-2.568(14)	9
Y^{3+}	d	1.806	0.634(18)	0.497(12)	-2.303(12)	9
Zr^{4+}	d	1.979	0.330(27)	0.288(21)	-2.268(21)	6
Nb^{5+}	d	1.945	0.426(30)	0.359(22)	-2.304(22)	7
Mo^{3+}	d	2.324	-0.237(0)	-0.268(0)	-2.057(0)	2
Mo^{5+}	d	1.919	0.415(46)	0.351(34)	-2.270(34)	4
Mo^{6+}	d	1.948	0.368(19)	0.317(14)	-2.265(14)	5
Pd^{2+}	d	1.402	0.918(0)	0.632(0)	-2.035(0)	2
Ag^{1+}	d	2.477	0.011(61)	0.011(60)	-2.489(60)	8
Ag^{2+}	d	1.778	0.548(0)	0.439(0)	-2.217(0)	2
Cd^{2+}	d	2.083	0.247(18)	0.223(15)	-2.306(15)	6
In^{3+}	p	2.065	0.162(34)	0.151(30)	-2.216(30)	5
Sn^{2+}	p	1.903	0.607(45)	0.484(30)	-2.387(30)	4
Sn^{4+}	p	1.845	0.567(9)	0.455(6)	-2.300(6)	3
Sb^{3+}	p	2.434	-0.341(87)	-0.410(131)	-2.025(131)	5
Sb^{5+}	p	1.990	0.130(39)	0.122(35)	-2.112(35)	4

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TABLE S1 – continued from previous page

Species	Block	R_0^*	B^*	λ^*	λ_-^*	n_{CN}
Te ⁴⁺	<i>p</i>	2.023	0.562(151)	0.458(104)	-2.481(104)	7
Te ⁶⁺	<i>p</i>	1.937	0.370(0)	0.318(0)	-2.254(0)	2
I ⁵⁺	<i>p</i>	2.077	0.476(34)	0.399(24)	-2.476(24)	6
I ⁷⁺	<i>p</i>	1.964	0.324(4)	0.283(3)	-2.247(3)	3
Cs ¹⁺	<i>s</i>	2.921	0.185(21)	0.175(19)	-3.096(19)	11
Ba ²⁺	<i>s</i>	2.453	0.308(11)	0.277(9)	-2.730(9)	10
La ³⁺	<i>f</i>	2.171	0.404(41)	0.348(31)	-2.519(31)	12
Ce ³⁺	<i>f</i>	2.142	0.360(50)	0.314(39)	-2.456(39)	9
Pr ³⁺	<i>f</i>	2.112	0.427(30)	0.364(22)	-2.476(22)	7
Nd ³⁺	<i>f</i>	2.307	0.207(50)	0.191(43)	-2.498(43)	8
Sm ³⁺	<i>f</i>	1.971	0.522(13)	0.429(9)	-2.400(9)	8
Eu ³⁺	<i>f</i>	2.088	0.396(28)	0.340(21)	-2.428(21)	6
Gd ³⁺	<i>f</i>	1.784	0.688(28)	0.530(17)	-2.314(17)	6
Tb ³⁺	<i>f</i>	1.862	0.598(18)	0.476(12)	-2.338(12)	5
Dy ³⁺	<i>f</i>	1.938	0.486(18)	0.403(13)	-2.341(13)	5
Ho ³⁺	<i>f</i>	1.967	0.444(16)	0.373(11)	-2.340(11)	5
Er ³⁺	<i>f</i>	1.854	0.564(10)	0.453(7)	-2.307(7)	5
Tm ³⁺	<i>f</i>	1.946	0.436(18)	0.367(13)	-2.313(13)	4
Yb ³⁺	<i>f</i>	1.995	0.372(7)	0.321(5)	-2.316(5)	4
Lu ³⁺	<i>f</i>	1.863	0.515(13)	0.420(9)	-2.283(9)	5
Hf ⁴⁺	<i>d</i>	1.929	0.362(16)	0.311(12)	-2.240(12)	4
Ta ⁵⁺	<i>d</i>	1.924	0.400(14)	0.340(10)	-2.265(10)	5
W ⁶⁺	<i>d</i>	1.936	0.305(20)	0.268(15)	-2.205(15)	4
Re ⁶⁺	<i>d</i>	1.928	0.363(0)	0.312(0)	-2.241(0)	2
Re ⁷⁺	<i>d</i>	1.948	0.377(72)	0.323(54)	-2.271(54)	4
Ir ⁴⁺	<i>d</i>	1.984	0.133(0)	0.125(0)	-2.109(0)	2
Pt ²⁺	<i>d</i>	1.785	0.370(0)	0.315(0)	-2.100(0)	2
Au ³⁺	<i>d</i>	1.924	0.363(13)	0.312(10)	-2.237(10)	3
Hg ¹⁺	<i>d</i>	2.230	0.233(28)	0.213(24)	-2.443(24)	4
Hg ²⁺	<i>d</i>	1.929	0.605(122)	0.484(81)	-2.413(81)	6
Tl ¹⁺	<i>p</i>	2.459	0.272(19)	0.247(16)	-2.705(16)	10
Tl ³⁺	<i>p</i>	2.054	0.469(42)	0.394(30)	-2.448(30)	4
Pb ²⁺	<i>p</i>	2.240	0.348(31)	0.306(24)	-2.547(24)	10
Pb ⁴⁺	<i>p</i>	2.080	0.353(23)	0.308(17)	-2.387(17)	3
Bi ³⁺	<i>p</i>	2.104	0.430(31)	0.366(23)	-2.470(23)	8
Th ⁴⁺	<i>f</i>	2.128	0.459(26)	0.388(19)	-2.516(19)	5
U ⁴⁺	<i>f</i>	1.938	0.787(91)	0.600(56)	-2.539(56)	3
U ⁵⁺	<i>f</i>	2.089	0.521(23)	0.431(16)	-2.521(16)	3
U ⁶⁺	<i>f</i>	2.080	0.603(92)	0.489(63)	-2.569(63)	7

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